

Discussion

In this experiment, a PIC16F628A microcontroller was used to create a simple LED chaser circuit. Seven LEDs were connected to PORTB, and the program was written to turn ON one LED at a time, creating a Knight Rider effect.

The PORTB pins were set as outputs using `TRISB = 0x00`, and the first LED was turned ON using `PORTB = 0x01`. The LED movement was controlled using shift operations. The `<<` operator moved the light from RB0 to RB6, while the `>>` operator moved it back from RB6 to RB0.

A delay was added using `Delay_ms(100)` so that the LED movement could be clearly seen. The comparator module was also disabled using `CMCON = 0x07` to make sure the pins worked properly as digital I/O pins.

During the experiment, it was important to connect the LEDs to the correct PORTB pins and use resistors to protect the LEDs. Overall, the circuit worked as expected and helped to understand how a microcontroller can control external devices using simple C programming.